

Towards the characterization of human settlement structure dynamics using the Global Human Settlement Layer and time series of landscape metrics

A study from Poland, 1975-2020

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Motivation and rationale

- Urban land cover and form are one of the most animated topics in the contemporary discussion on global urbanisation (e.g. Arribas-Bel and Fleischmann, 2022; Taubenböck, 2025).
- A growing body of work on landscape metrics, measuring landscape characteristics that describe size, density, shape and distribution of urban areas (e.g. Yang et al., 2022; Chakraborty et al., 2024).

Motivation and rationale

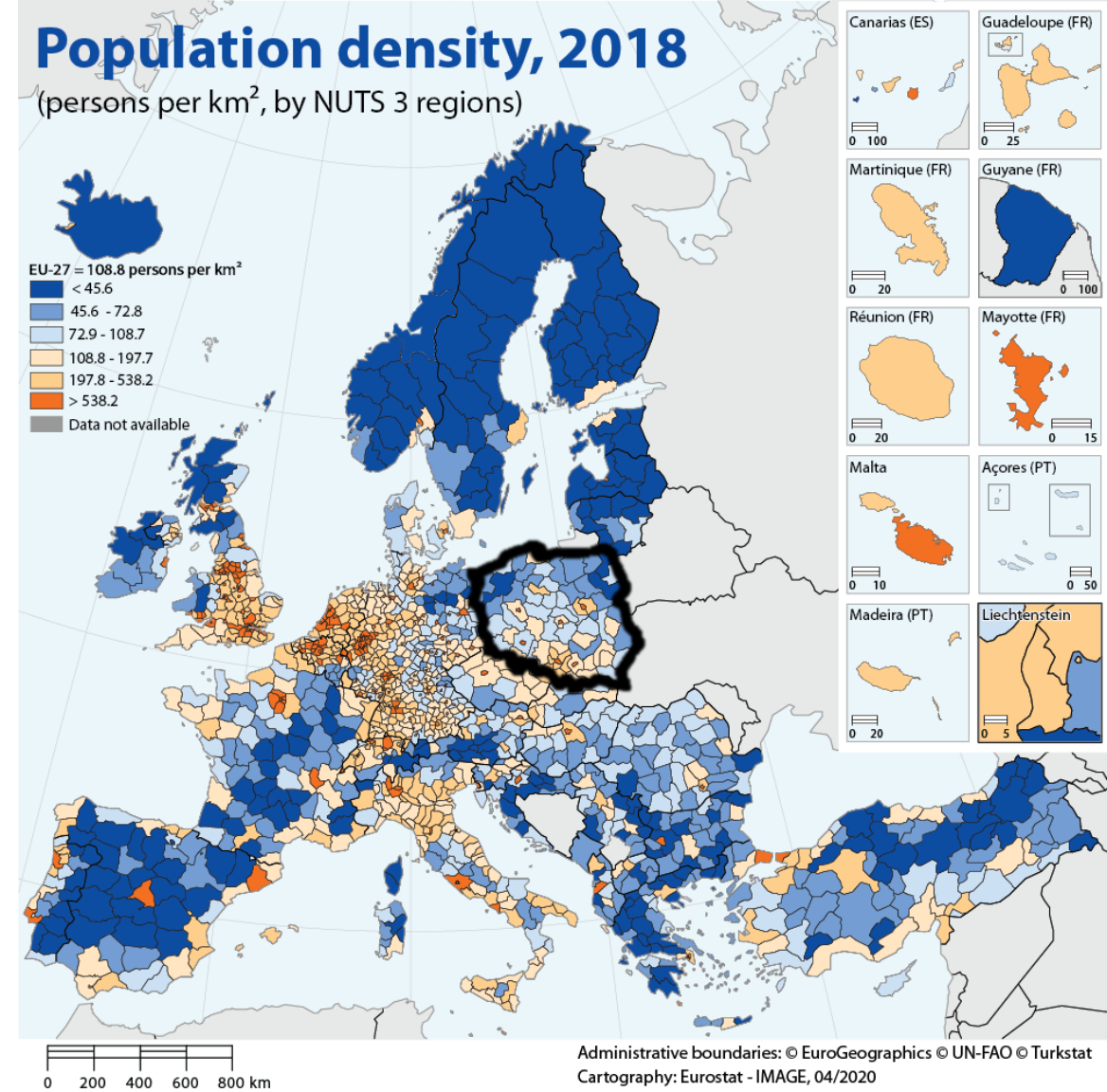
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- A growing body of work on landscape metrics, measuring landscape characteristics that describe size, density, shape and distribution of urban areas (e.g. Yang et al., 2022; Chakraborty et al., 2024).
- The increasing availability of global multitemporal data describing the human presence on Earth, e.g. Global Human Settlement Layer data (Pesaresi et al., 2024), World Settlement Footprint (WSF) Evolution data (Marconcini et al., 2021)...
- Need for novel analytical frameworks that leverage thematic richness of data.

Towards the characterization of human settlement structure dynamics

- Goal: Characterize spatial and temporal changes in human settlements using global, multitemporal, and multidimensional data.
- Data: Global Human Settlement Layer (GHSL) – built-up surface and population data (100 m, 1975–2020).
- **Method: Quantile-based classification and cross-tabulation of built-up surface and population densities at Local Administrative Unit (LAU) level.**
- Analysis: We track changes over time and apply landscape metrics to describe spatial structure and dynamics.
- Outcome: Integrated framework for large-scale, consistent monitoring of settlement transformation.

Case study of Poland

- Area of ~300 000 km²
- Population in 2025 of ~38 million people
- 2 477 municipalities (LAU level 2 units)

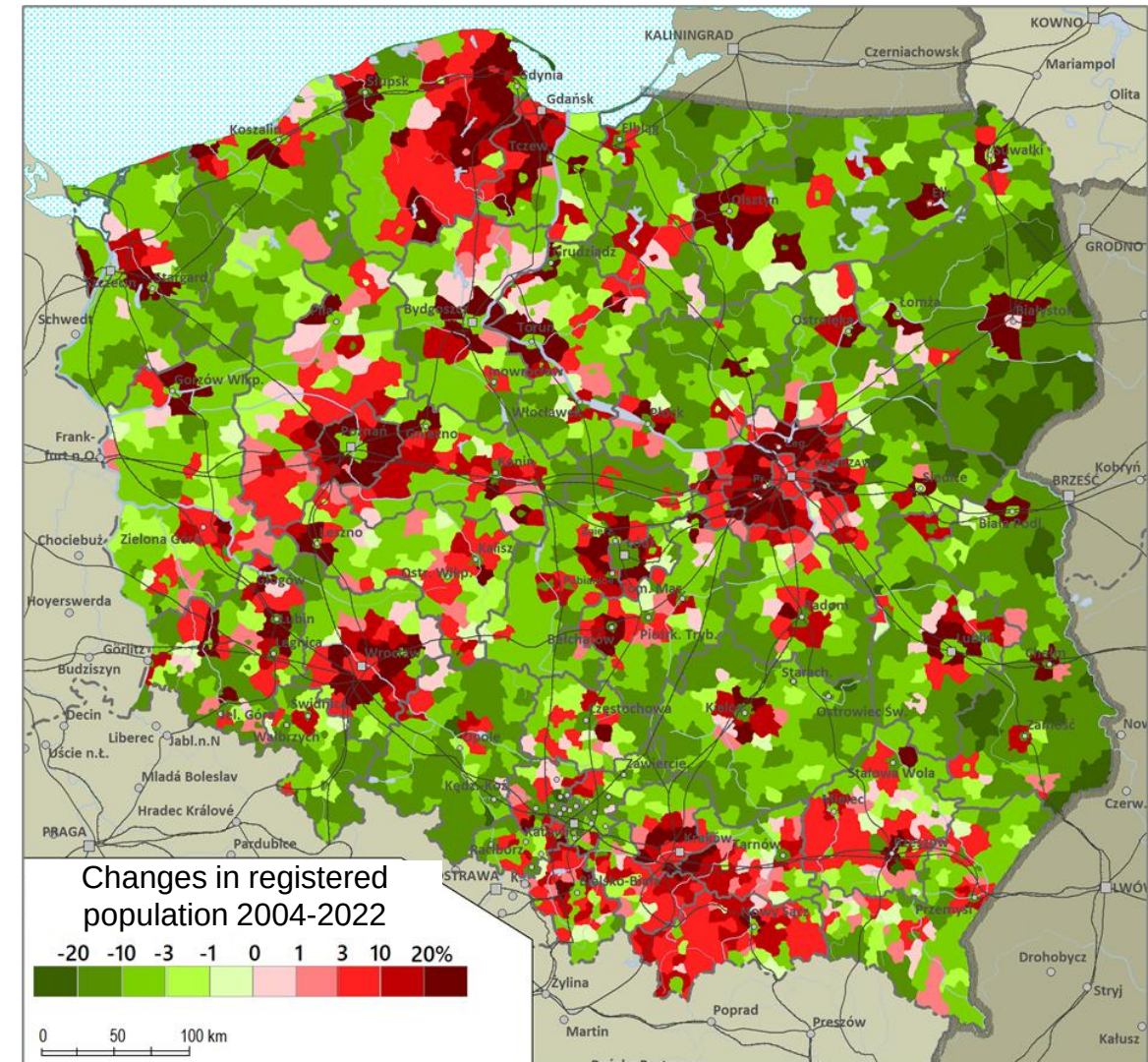


ec.europa.eu/eurostat 

Population density map from: <https://ec.europa.eu/eurostat/>

Major socio-economic changes between 1975 and 2020

- after 1980: socio-economic crisis and slowdown in housing construction despite the demographic peak (1975–1985)
- **1989: transition from a centrally planned economy to a market economy**
- 2003: reform of local spatial planning
- 2004: accession to the European Union (EU), rapid development of the economy and construction infrastructure
 - intensive suburbanisation on the wave of another demographic peak

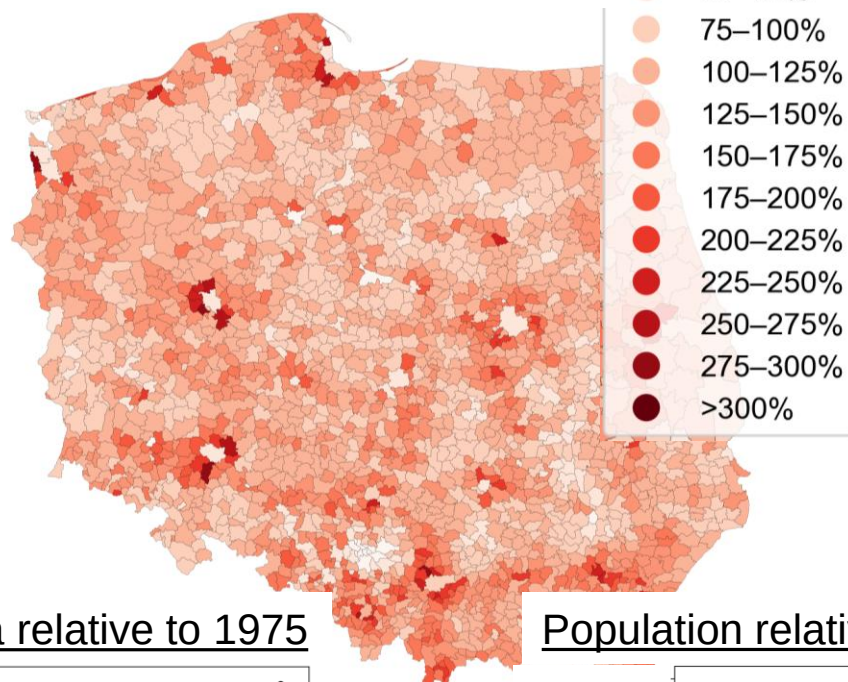


From: Śleszyński P. (2022)

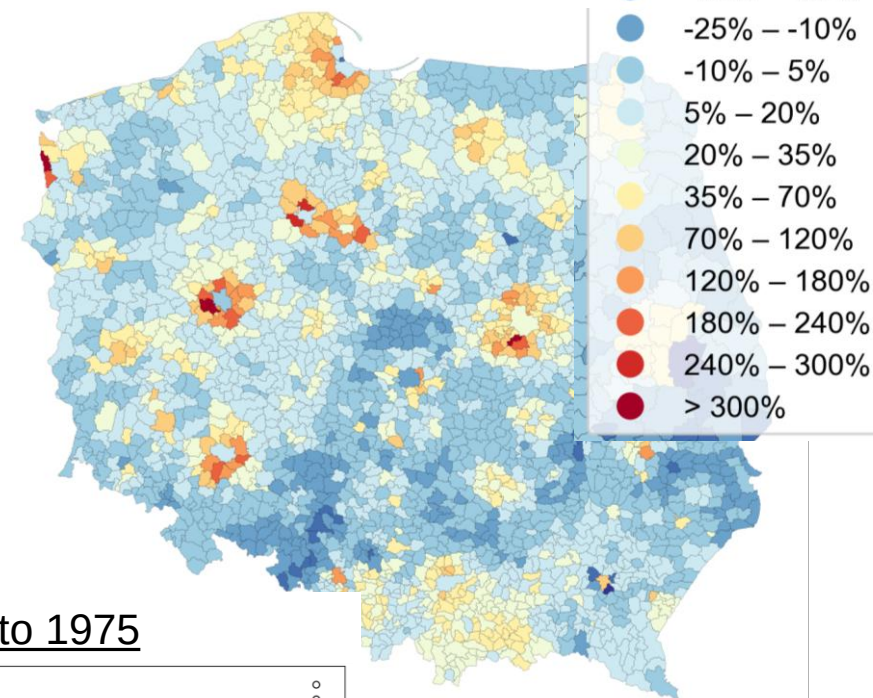
Change in built-up surface area and population aggregated at LAU level

- Increase in built-up surface area
- Redistribution of population

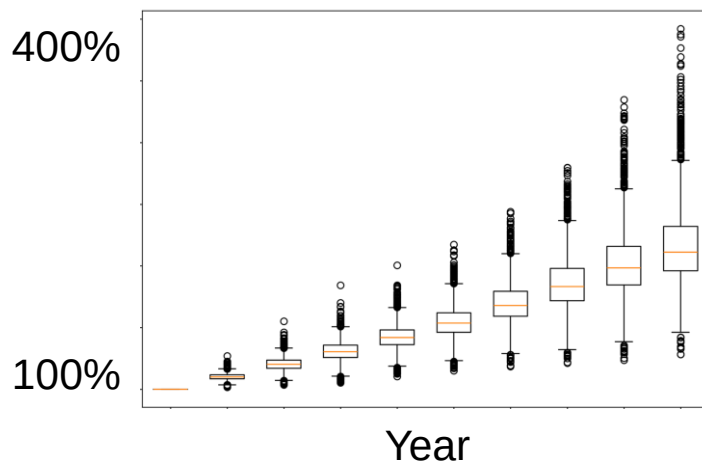
Change in built-up surface area
1975-2020



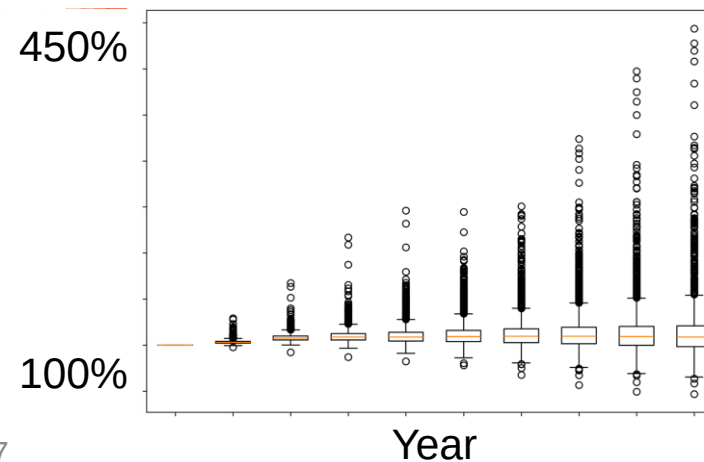
Change in population density
1975-2020



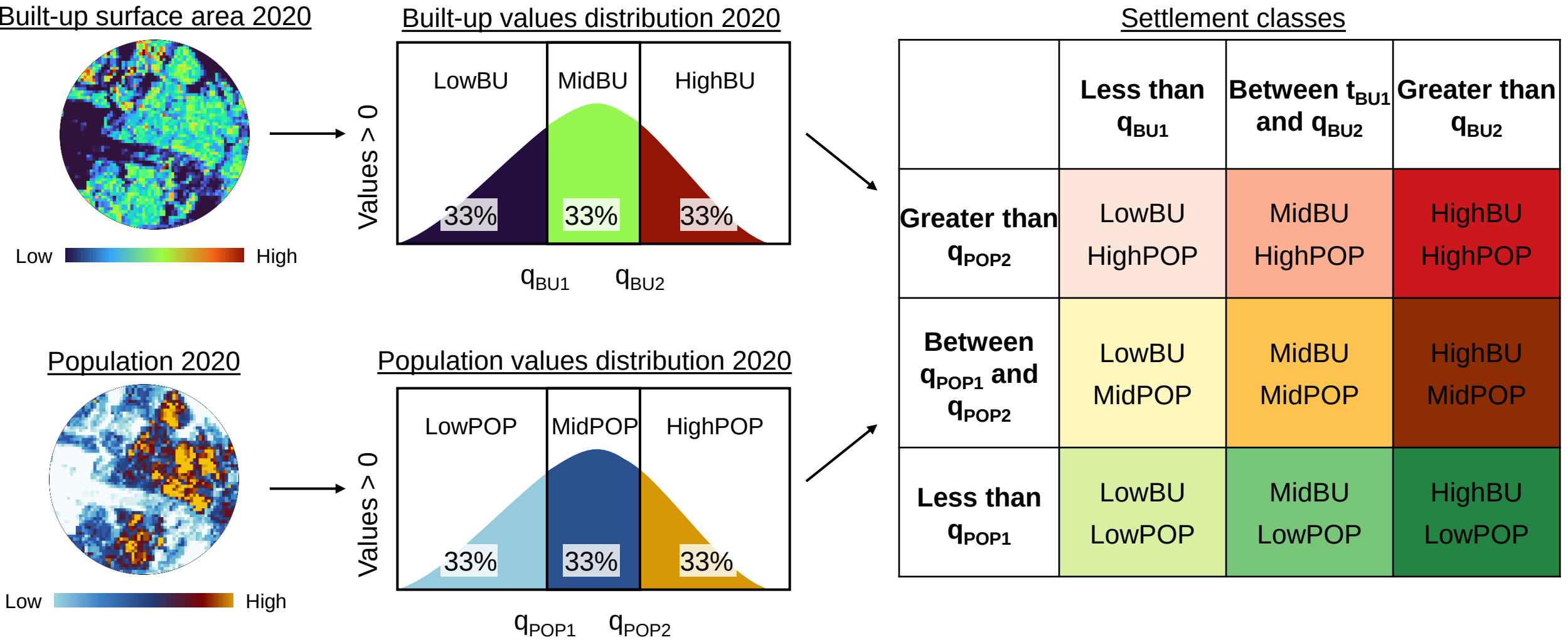
Built-up surface area relative to 1975



Population relative to 1975

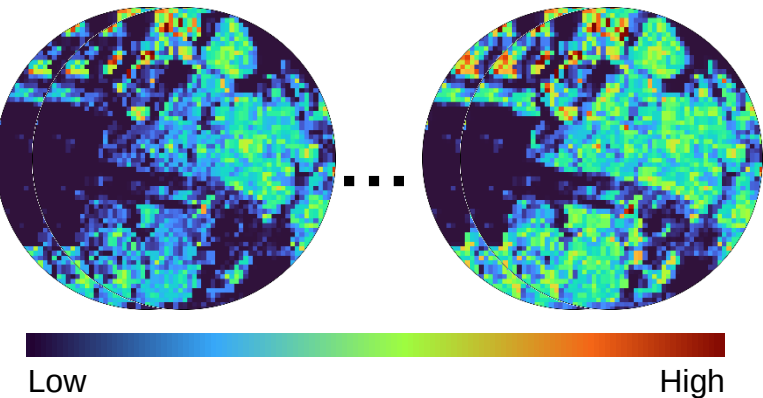


Defining quantile-based settlement classification for AOI

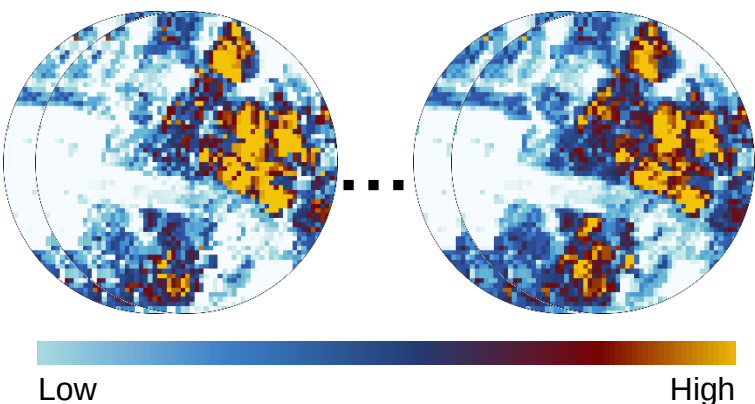


Multi-temporal classification of settlements

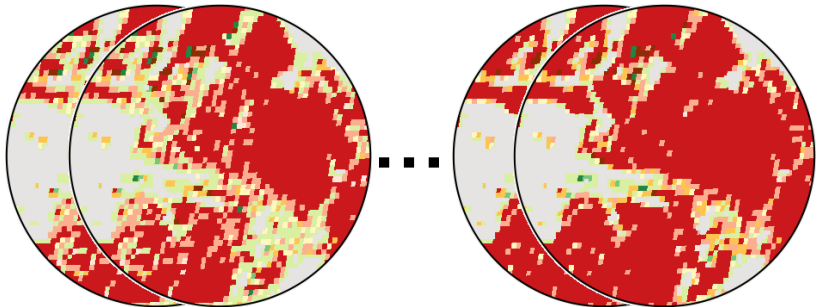
Built-up surface area from 1975 to 2020



Population from 1975 to 2020



Settlement classification from 1975 to 2020



	Less than q_{BU1}	Between t_{BU1} and q_{BU2}	Greater than q_{BU2}
Greater than q_{POP2}	LowBU HighPOP	MidBU HighPOP	HighBU HighPOP
Between q_{POP1} and q_{POP2}	LowBU MidPOP	MidBU MidPOP	HighBU MidPOP
Less than q_{POP1}	LowBU LowPOP	MidBU LowPOP	HighBU LowPOP

q_{BU1} , q_{BU2} : Built-up surface area tertiles for 2020 q_{POP1} , q_{POP2} : Population tertiles for 2020

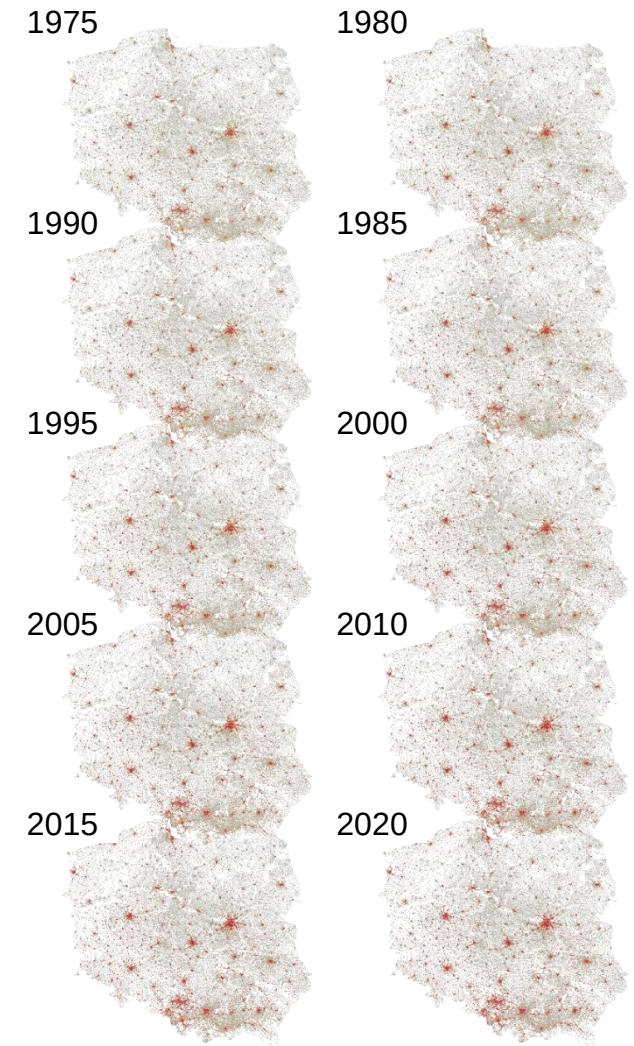
Settlement classification for Poland

Grid cell value: Built-up surface area

Grid cell value: Population		Grid cell value: Built-up surface area		
		Less than q_{BU1} ($\leq 2\%$ / ha)	Between q_{BU1} and q_{BU2} (2 – 9% / ha)	Greater than q_{BU2} ($> 9\%$ / ha)
	Greater than q_{POP2} (> 500 p. / km ²)	LowBU HighPOP	MidBU HighPOP	HighBU HighPOP
	Between q_{POP1} and q_{POP2} (100 – 500 p. / km ²)	LowBU MidPOP	MidBU MidPOP	HighBU MidPOP
	Less than q_{POP1} (≤ 100 p. / km ²)	LowBU LowPOP	MidBU LowPOP	HighBU LowPOP

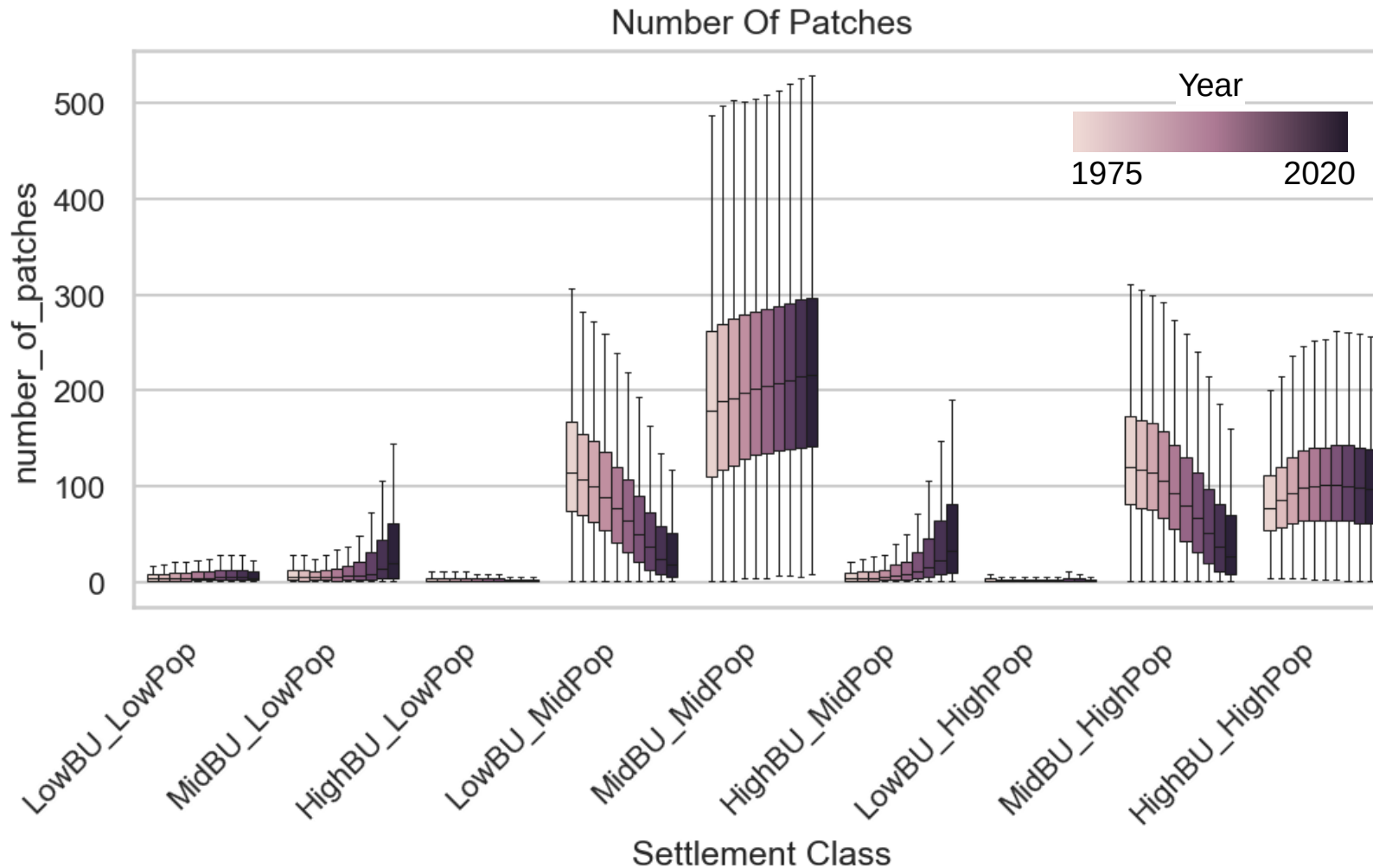
q_{BU1} , q_{BU2} : Built-up surface area tertiles for 2020

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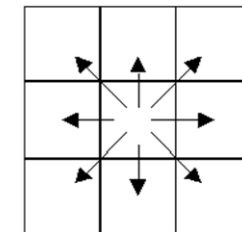
- The continuous gridded built-up surface area and population estimates are discretized into quantile-based classes.
- Based on that cross-tabulation, the area of interest (Poland) is categorized into nine settlement classes, for each time-step between 1975 and 2020.

Landscape metrics calculated on LAU level



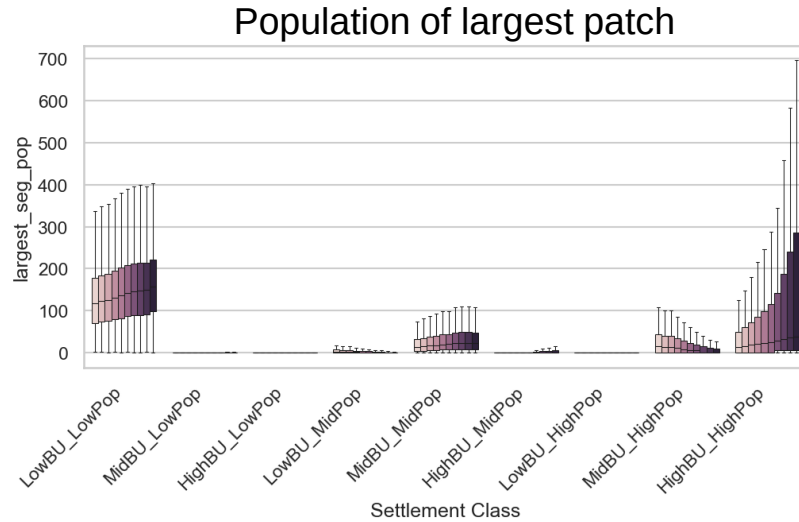
Number of polygons of given settlement class in LAU unit

- Landscape metrics were initially used to quantify shape and pattern of vegetation and were later applied to measure various aspects of landscape configuration and composition.
- They mostly rely on aerial and satellite imagery and use polygons of homogenous land-use (“patches”) as the key unit of analyses (Clifton et al., 2008; Reis, Silva and Pinho, 2015).



queen's case/Moore neighbourhood

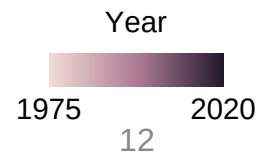
Landscape metrics combined with built-up surface and population data



Represents the population within the largest contiguous area of a given settlement class.

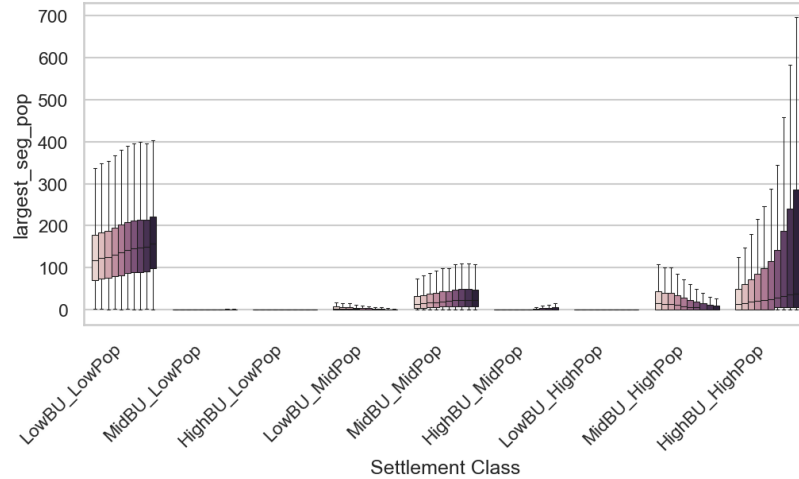
Suitable for:

- assessing changing settlement structure
- ranking areas by size at local and regional scales.



Landscape metrics combined with built-up surface and population data

Population of largest patch

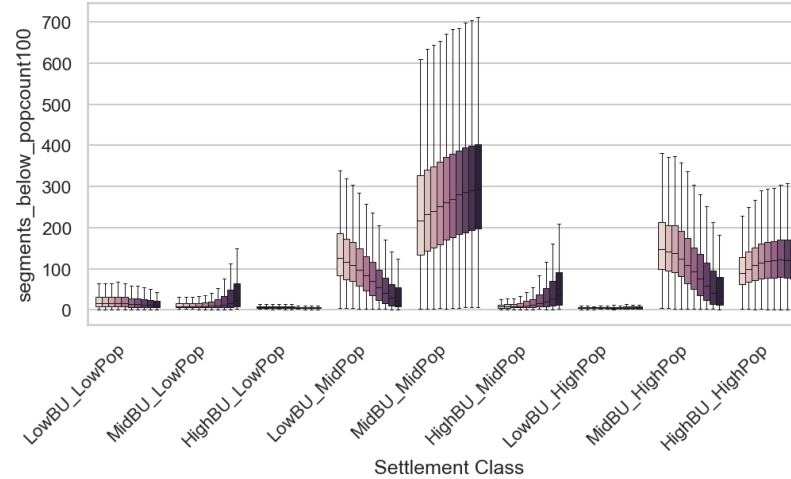


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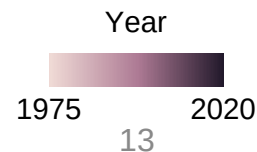
Suitable for:

- assessing changing settlement structure
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Number of patches with population < 100 inhabitants

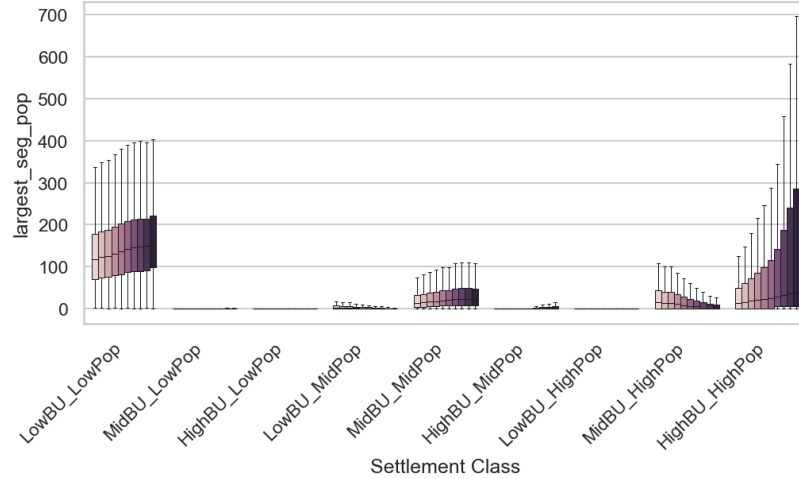


Counts small, sparsely populated settlement patches. A higher value indicates greater population dispersion within the area.



Landscape metrics combined with built-up surface and population data

Population of largest patch

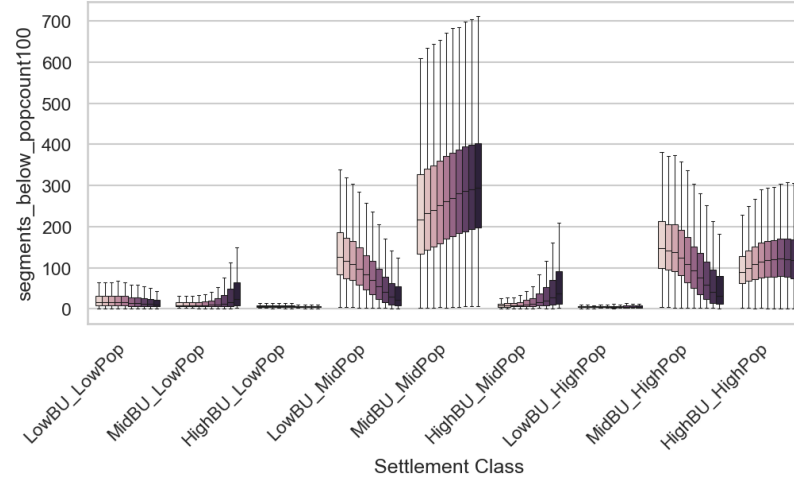


Represents the population within the largest contiguous area of a given settlement class.

Suitable for:

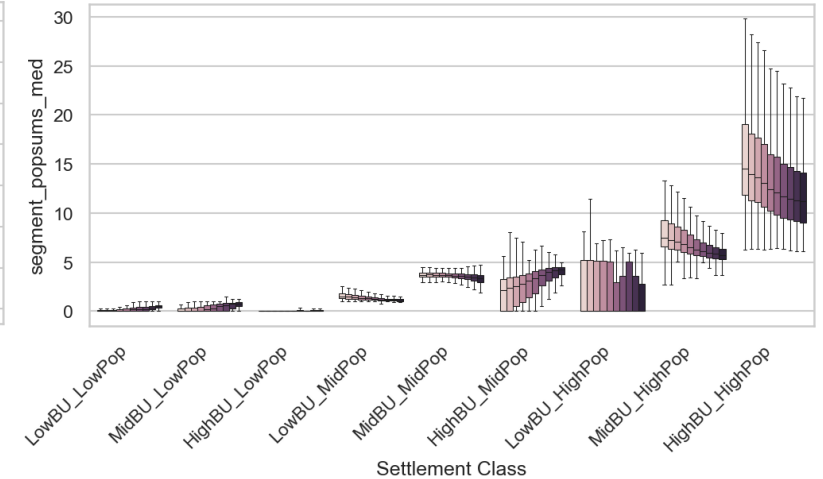
- assessing changing settlement structure
- ranking areas by size at local and regional scales.

Number of patches with population < 100 inhabitants

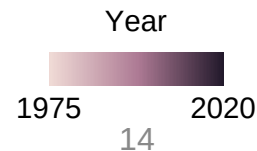


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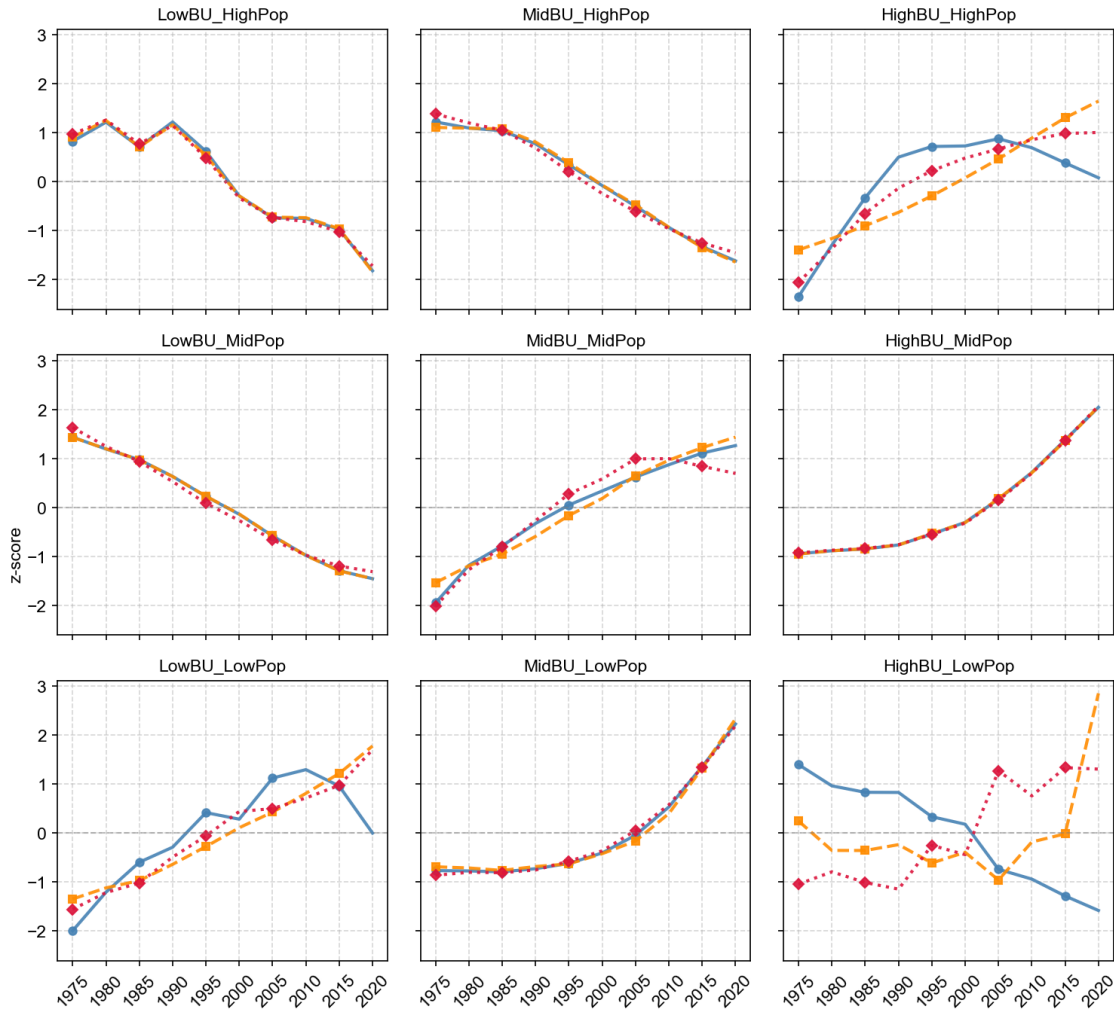
Median population in patches



Represents the typical population size per settlement patch. Useful for tracking depopulation trends and urban shrinkage over time.



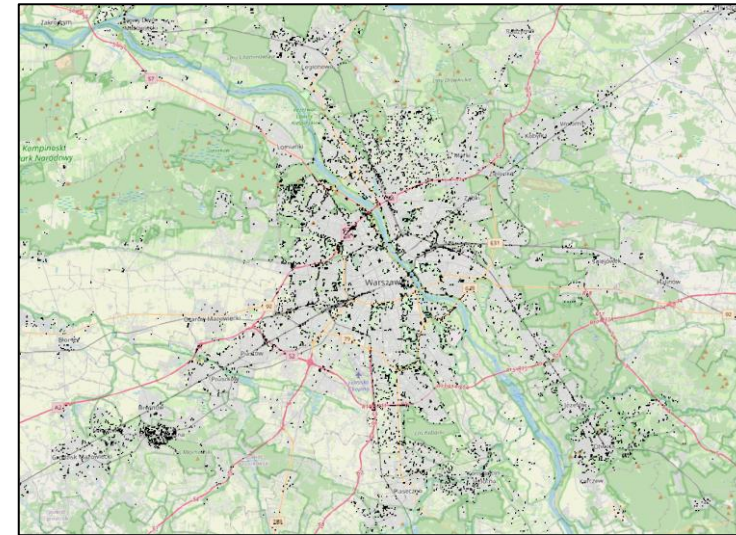
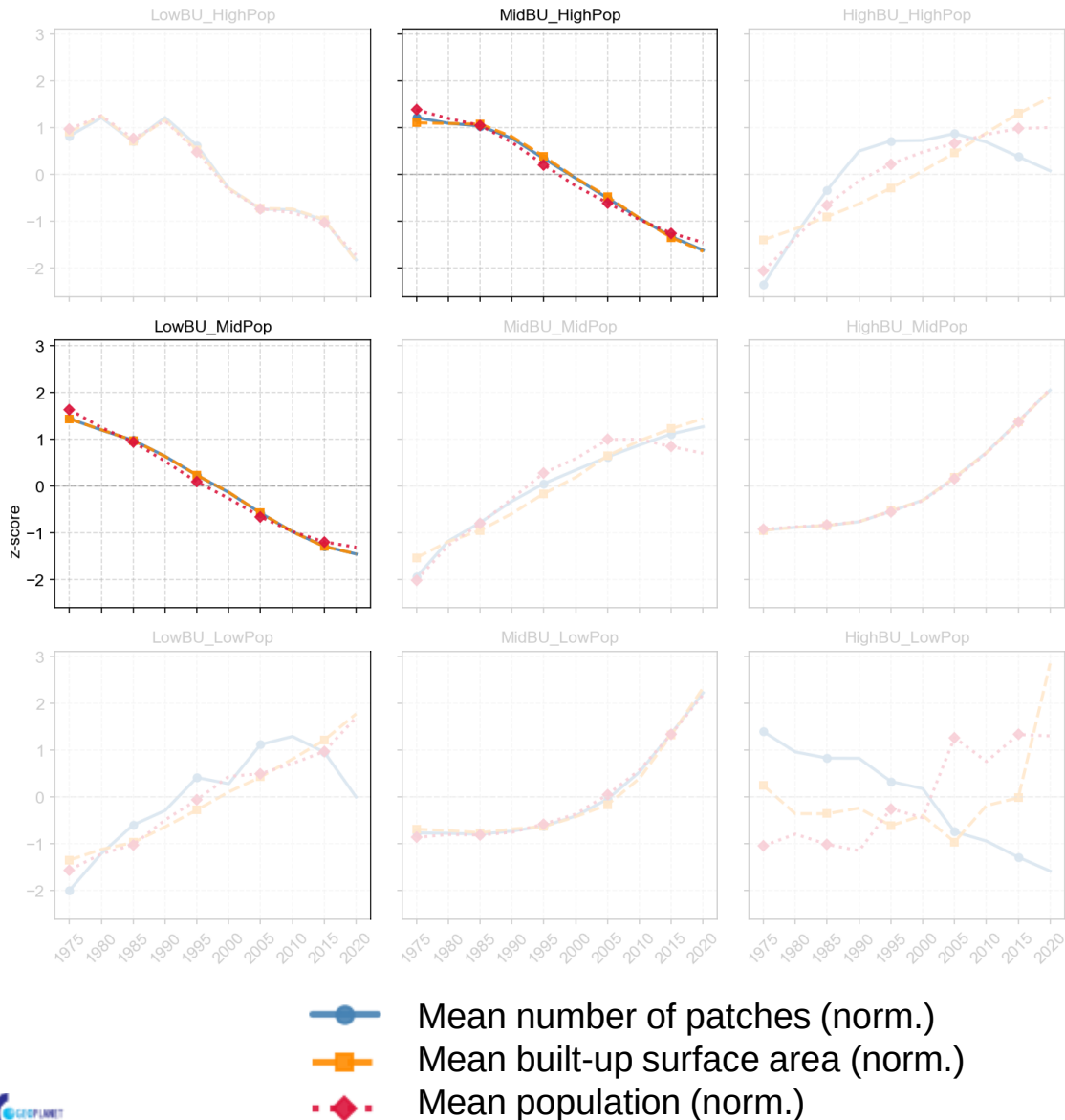
Integrated characterization of human settlements over time



- 2D feature space that considers quantitative and spatial changes in human settlements in time

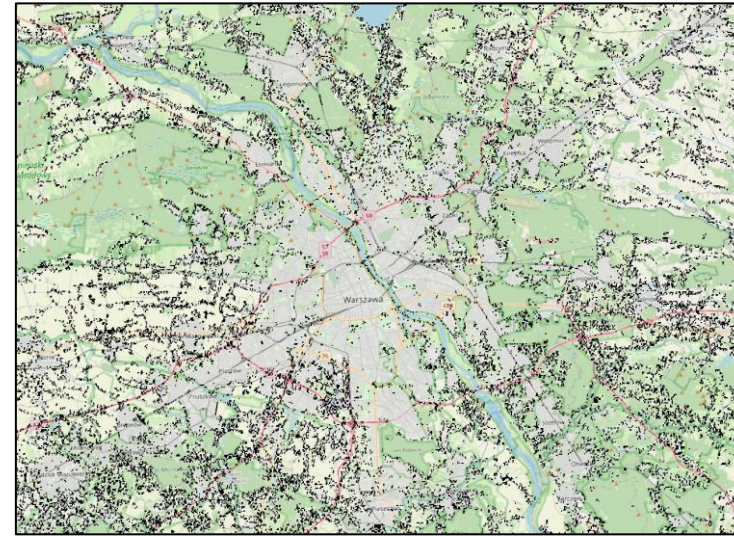
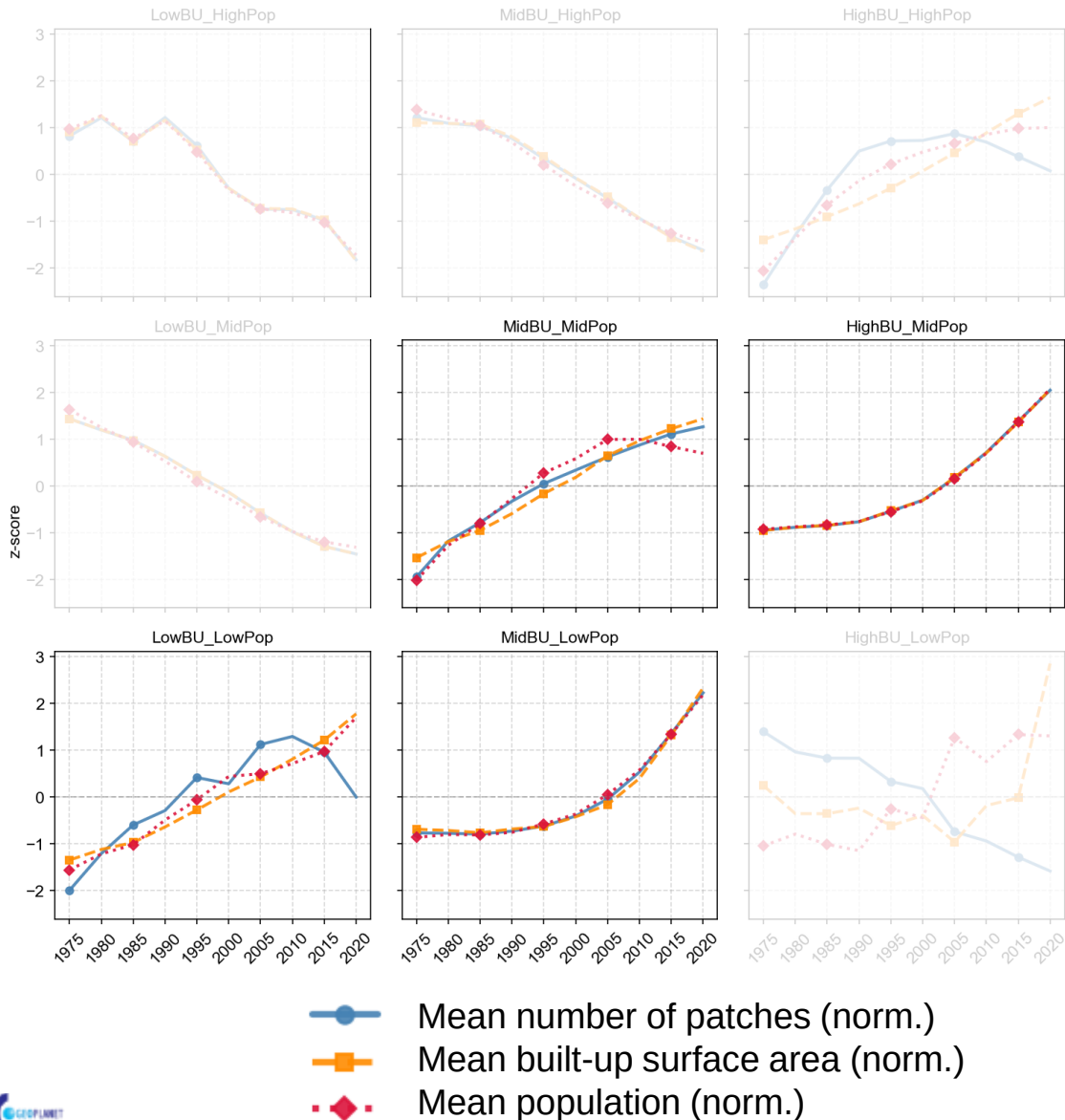
 Mean number of patches (norm.)
 Mean built-up surface area (norm.)
 Mean population (norm.)

Integrated characterization of human settlements over time



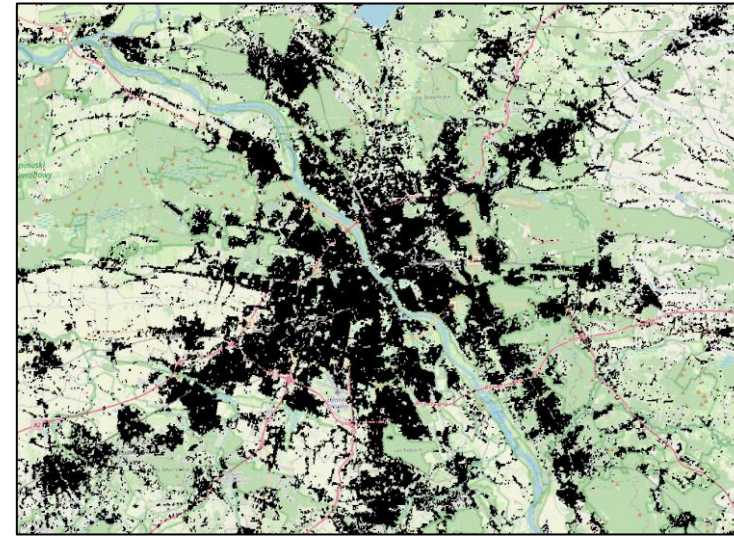
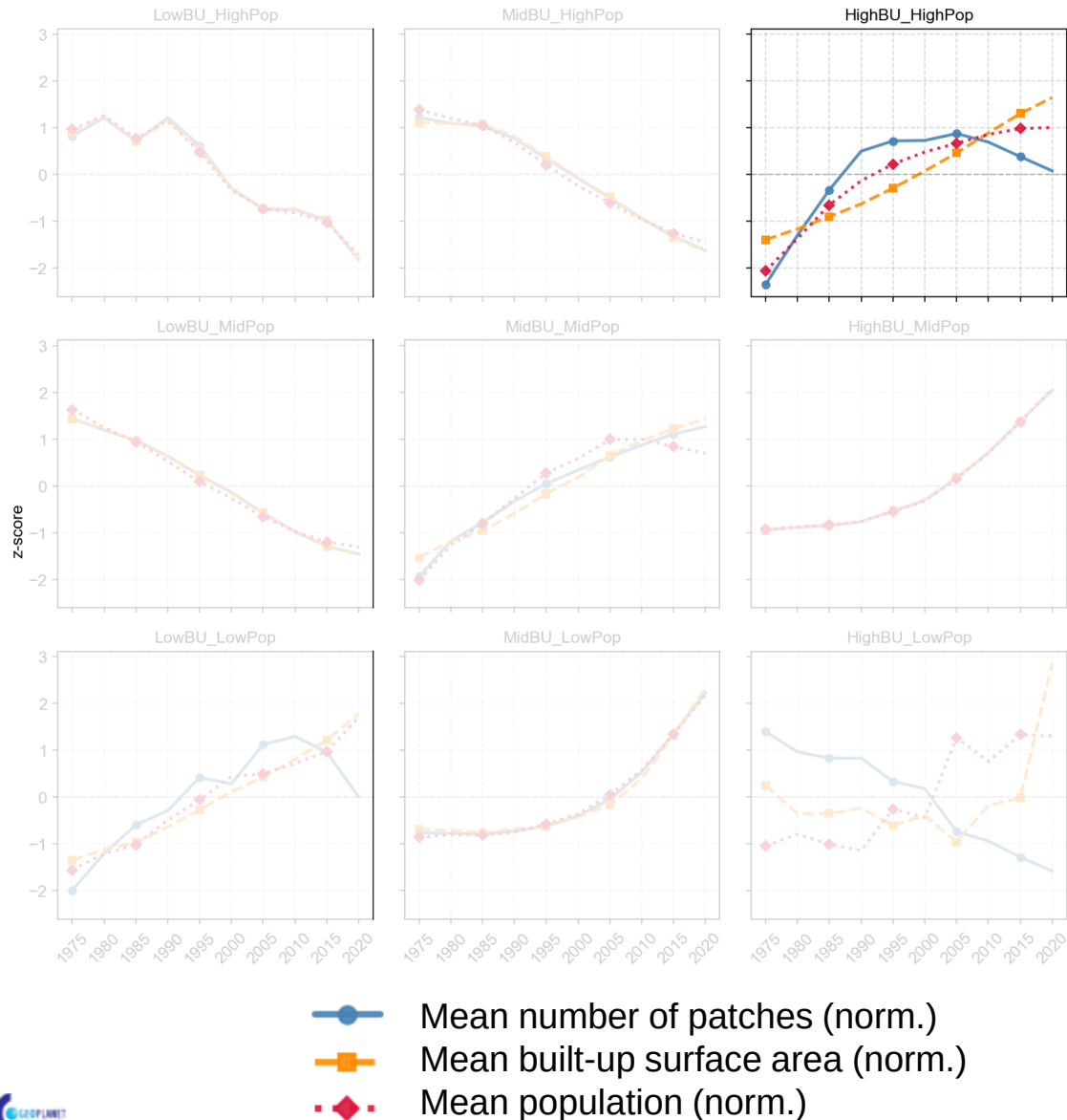
- Inner-urban areas were clustering and decreasing in size

Integrated characterization of human settlements over time



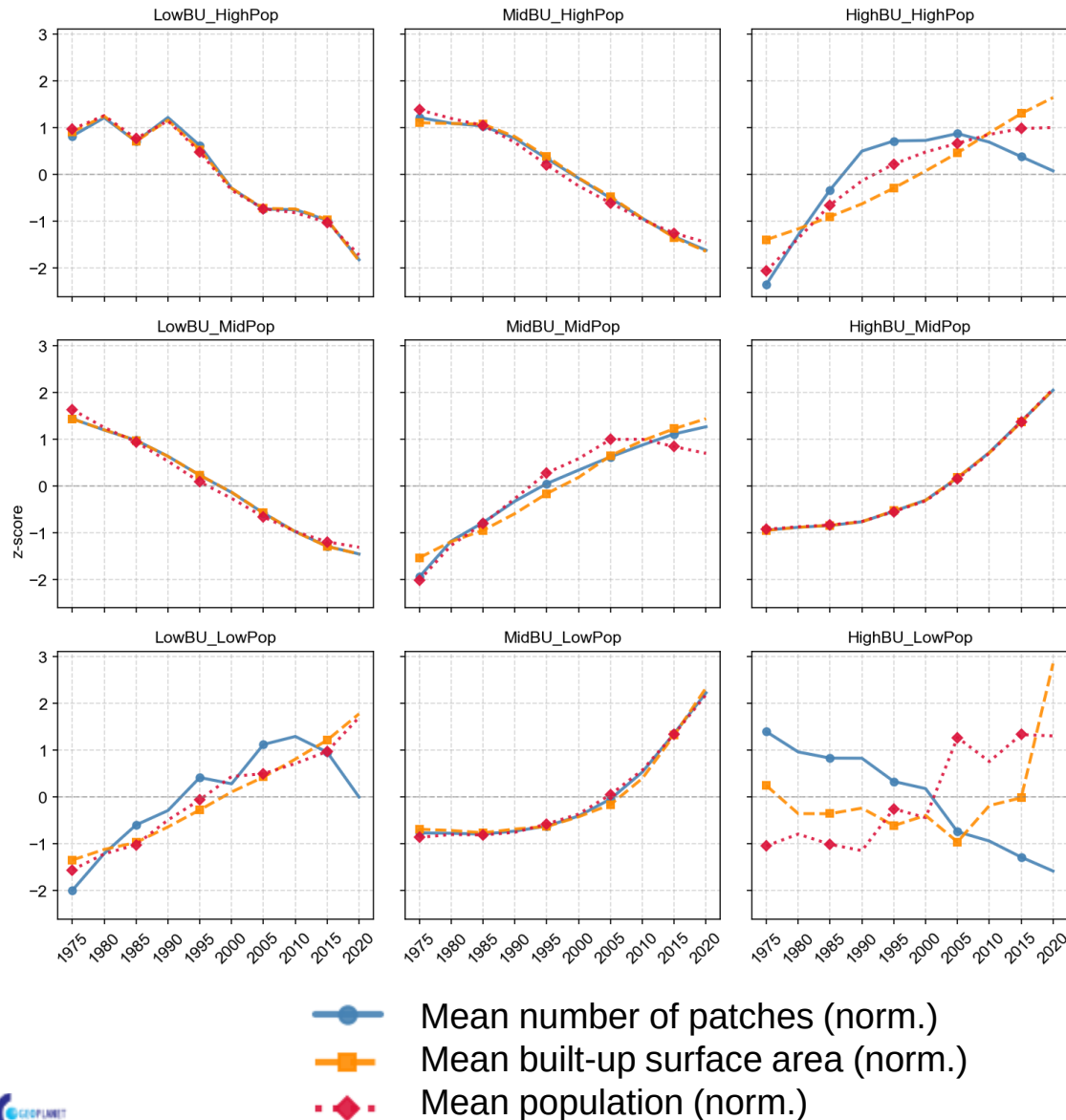
- Outer-urban areas were increasing in size and becoming more dispersed
- The process was accelerating in less densely populated areas

Integrated characterization of human settlements over time



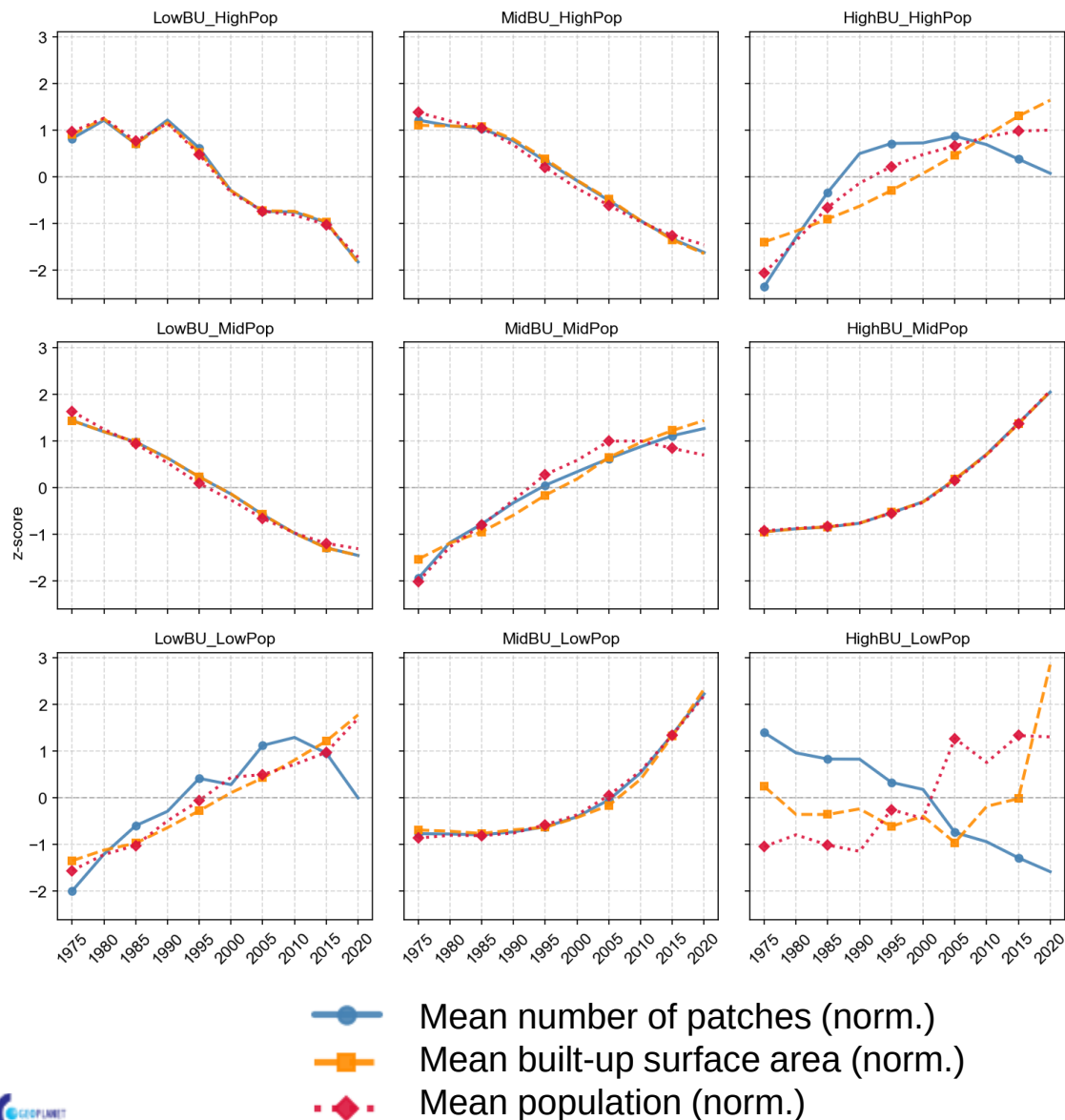
- Urbanized areas were increasing in size and becoming more clustered

Outlook: integrated framework



- The data-driven framework leverage open, global, multitemporal data on the built environment and population:
 - The global coverage of the data enables flexible selection of area of interest (suitable for country-level cross-comparison, peri-urban analyses, multi-level evaluation..)
 - Temporal extent of ~50 years allows for detailed characterization of human settlements over time.
 - Resolution of 100 m captures divergent settlement trends.

Outlook: future work



Opportunities:

- Assessment of changes in settlement types over time – important for explaining urbanisation models, in particular the stage of suburban development; explanation of population redistribution processes.
- The LAU level allows for linking the identified classes with socio-economic data.

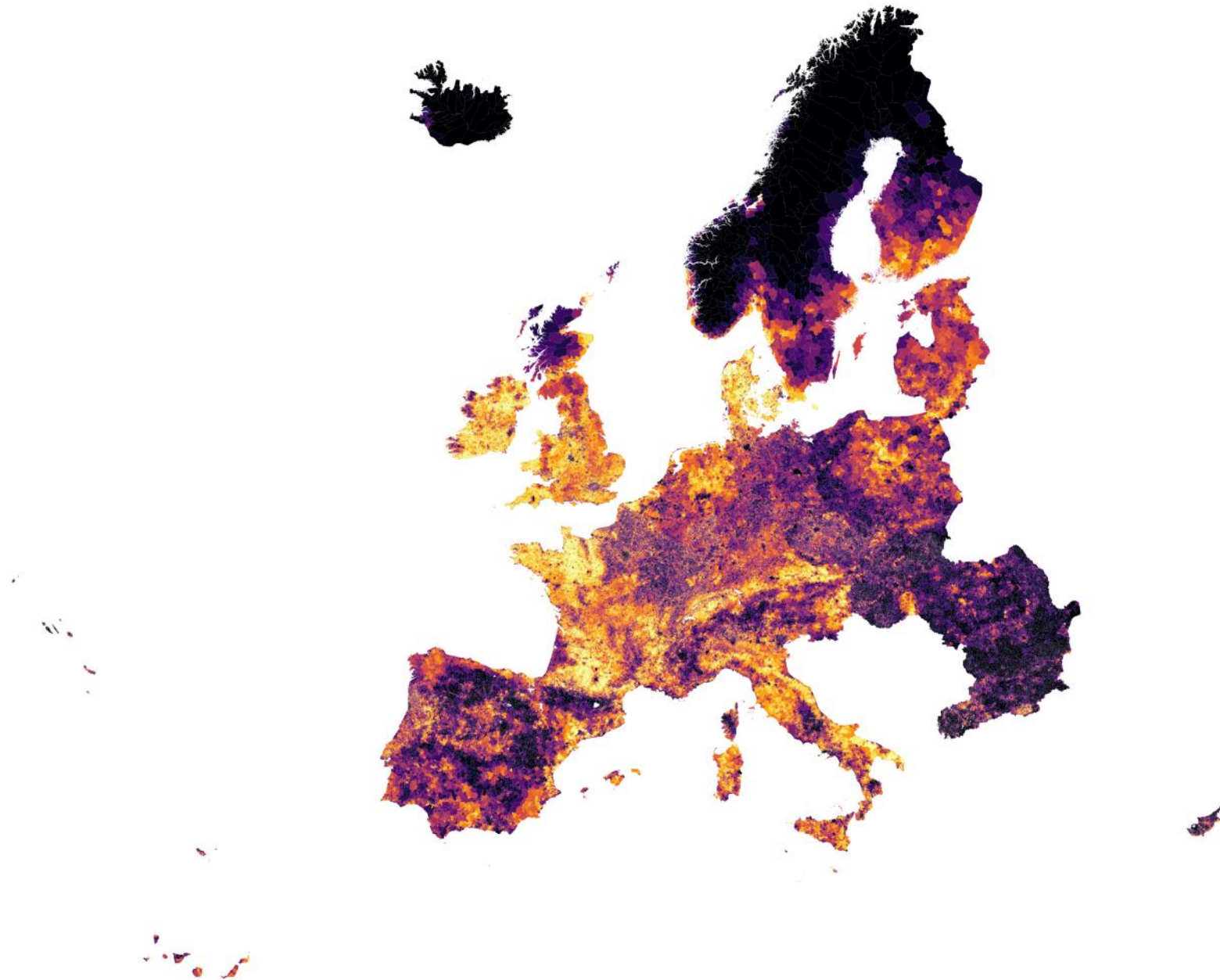
Open questions:

- How to disentangle the signals of re-classification vs structural change?
- How to include the depopulation of built-up areas?
- Choice of the number of settlement classes: some classes are underrepresented (HighBU_LowPOP)

We are currently working on expanding this framework on the EU level.

1975

Patch density



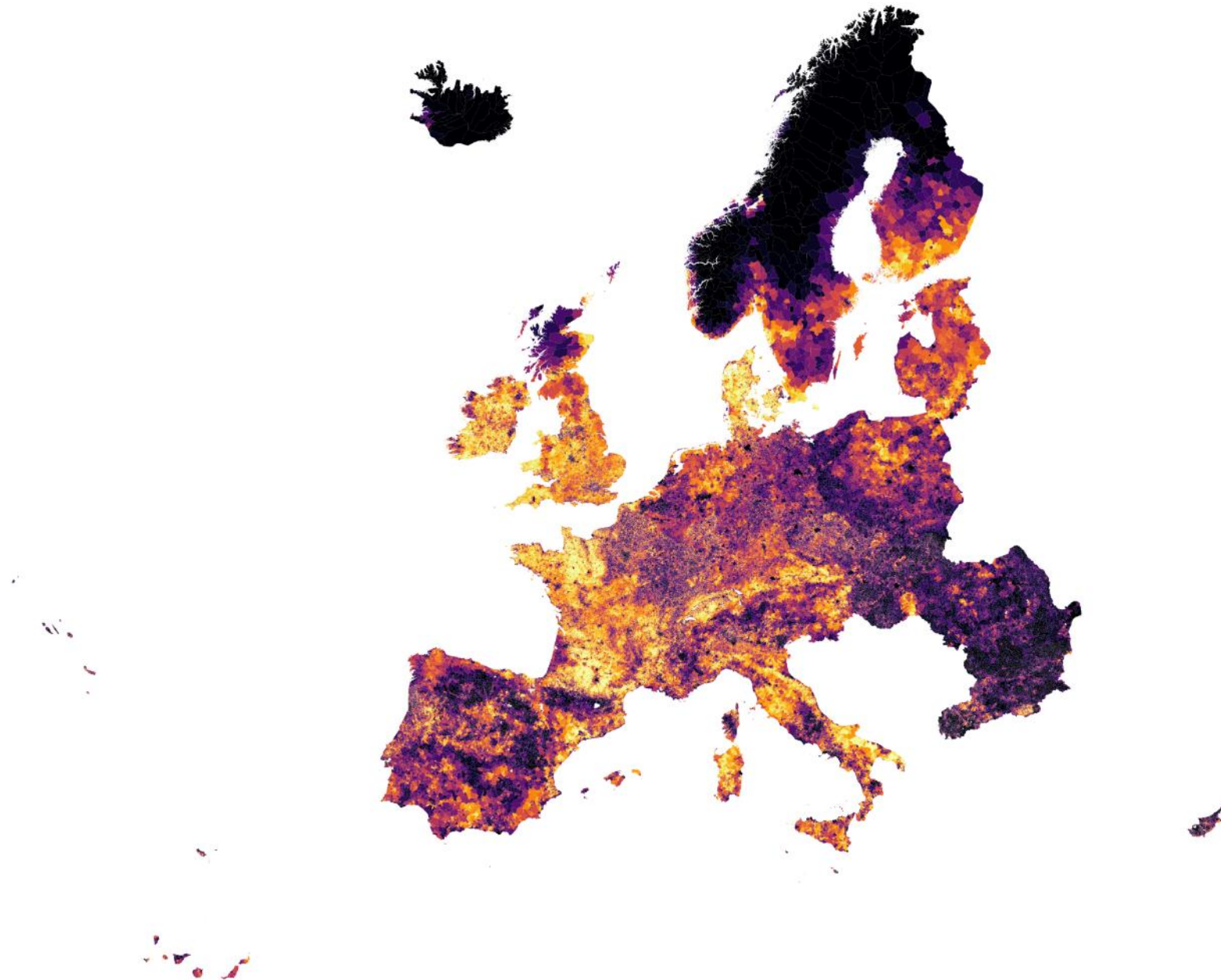
Norm. value

High

Low

2020

Patch density



Norm. value

High

Low

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Thank you!

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Comparative assessment on the EO-derived GHSL information

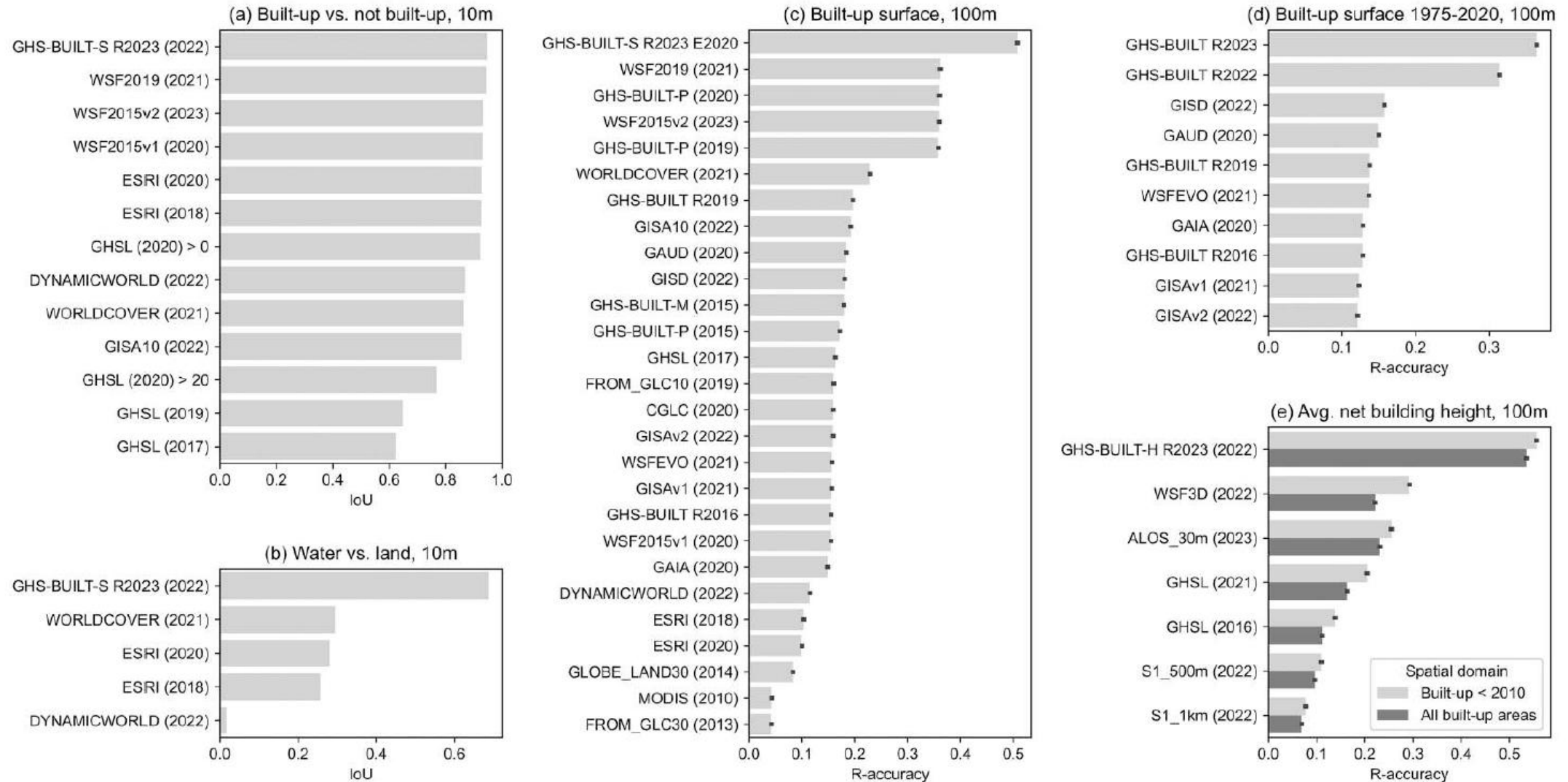
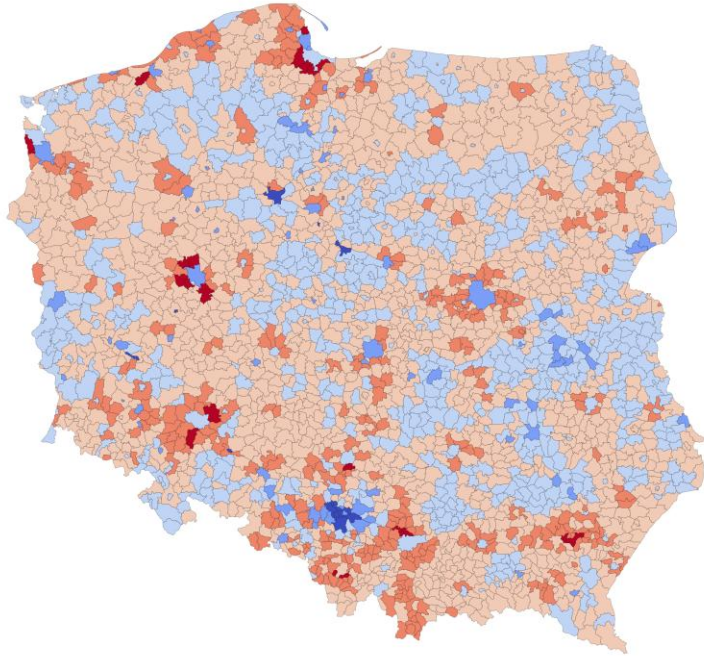


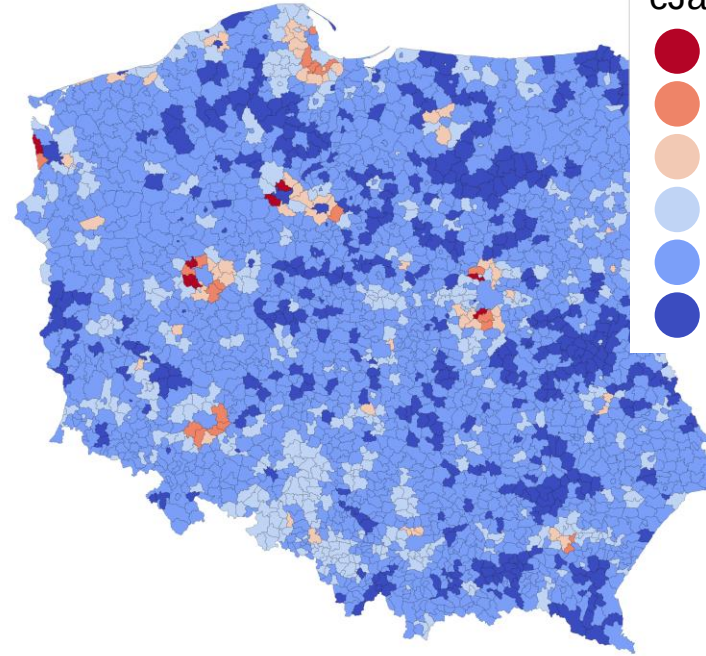
Fig. 25 from Pesaresi et al. (2024)

Grid level agreement between 1975 and 2020

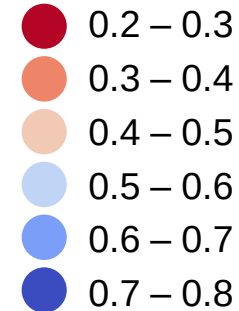
Built-up surface area



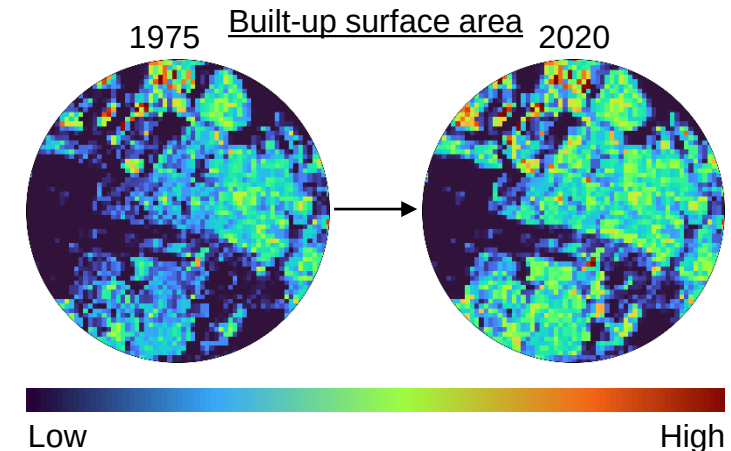
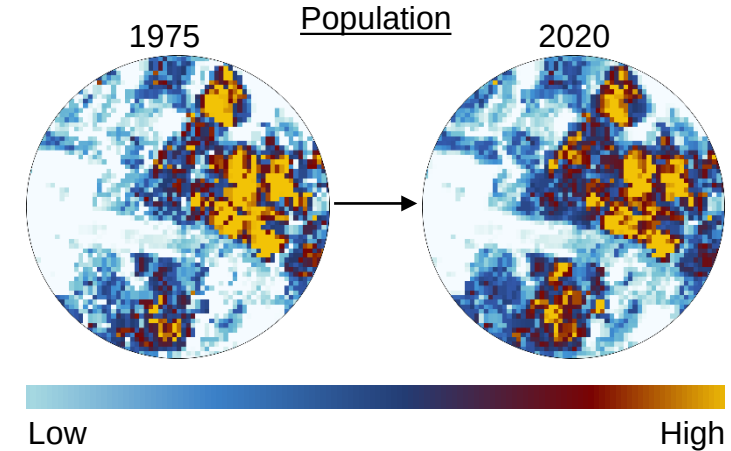
Population



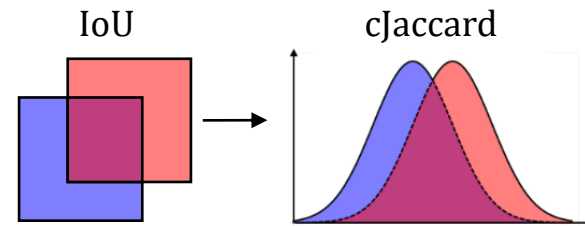
cJaccard [0-1]:



Gridded spatial raster datasets
in 100 m resolution:



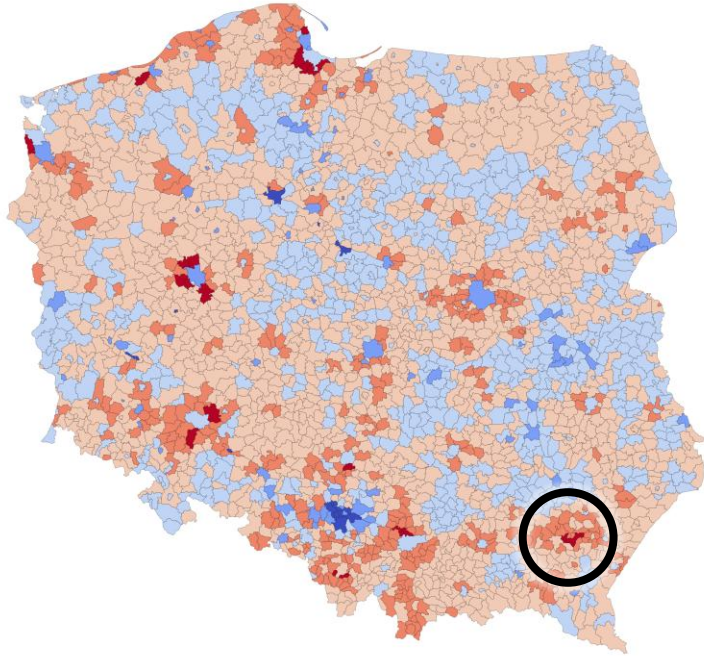
$$cJaccard = \frac{\sum_{i=1}^N \min(m_i, r_i)}{\sum_{i=1}^N \max(m_i, r_i)}$$



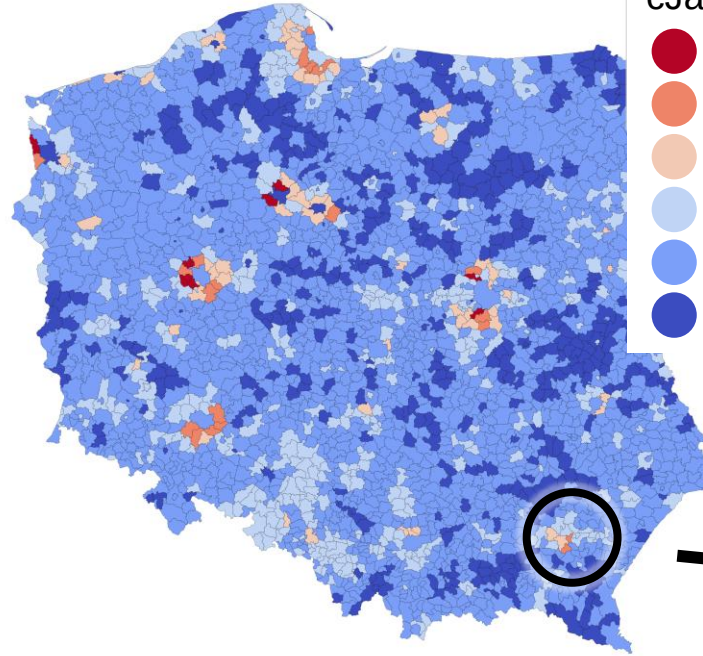
Krasnodębska et al., (2024) "Advancing Precision, Recall, F-score, and Jaccard Index: An approach for continuous gridded data." *Environmental Modelling and Software*

Grid level agreement between 1975 and 2020

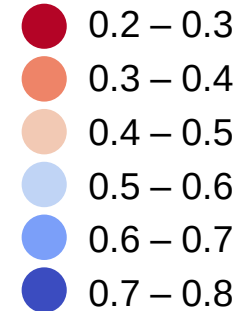
Built-up surface area



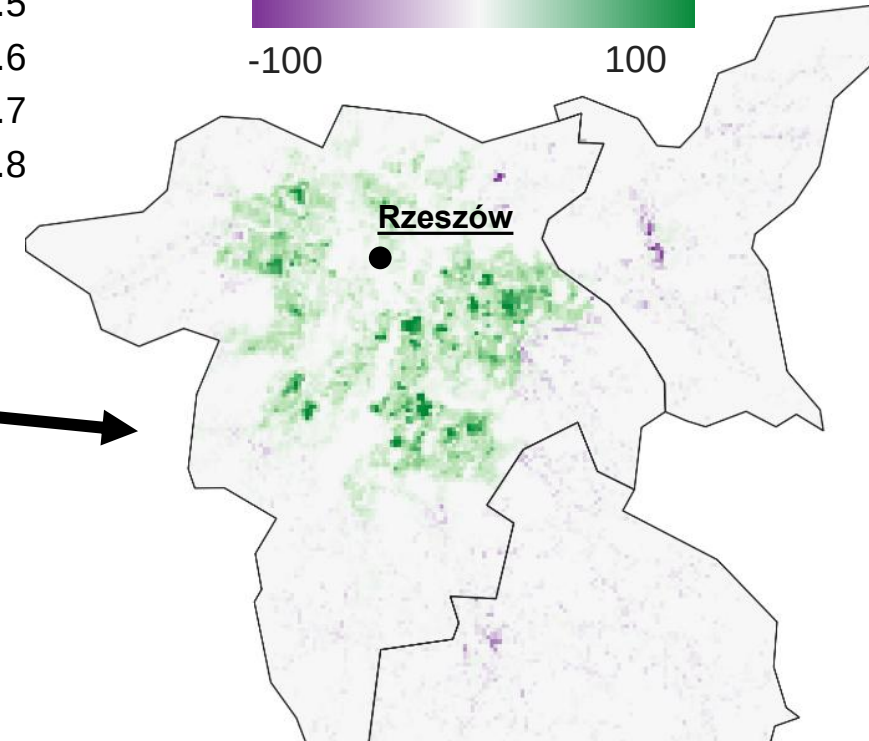
Population



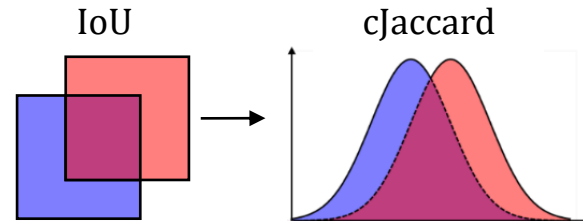
cJaccard [0-1]:



Difference in population
1975-2020



$$cJaccard = \frac{\sum_{i=1}^N \min(m_i, r_i)}{\sum_{i=1}^N \max(m_i, r_i)}$$



Krasnodebska et al., (2024) "Advancing Precision, Recall, F-score, and Jaccard Index: An approach for continuous gridded data." *Environmental Modelling and Software*

Simultaneous trends of built-up surface expansion, urban densification and suburban depopulation